UNDERGRADUATE THESIS

ENGLISH THESIS TITLE WRITTEN IN TITLE CASE

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SURABAYA
2026

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Declare that this undergraduate thesis does not contain parts of another scientific work submitted to obtain an academic degree at any higher education institution, nor does it contain work or opinions written or published by other persons/institutions, except those properly cited and listed in the bibliography.

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ABSTRACT

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Complete this section: Write the abstract in the selected document language. When `\ThesisLanguage` is set to `english`, labels and guidance become English.

The abstract summarizes the context, objective, method, data or research object, main findings, and conclusion. Avoid citations and unsupported claims.

Keywords: first keyword, second keyword, third keyword

ACKNOWLEDGEMENTS

The author expresses gratitude to God Almighty for the completion of this Undergraduate Thesis entitled **“English Thesis Title Written in Title Case”**. This undergraduate thesis is prepared as one of the requirements for obtaining the Bachelor of Computer Science degree at the Informatics Study Program, Faculty of Computer Science, Universitas Pembangunan Nasional Veteran Jawa Timur.

Writing tip: Adjust acknowledgements to people and institutions that actually contributed. Use formal language and verify names and titles.

1. Nama Dosen Pembimbing I, Gelar as Advisor I for guidance, direction, and time during the undergraduate thesis preparation.
2. Nama Dosen Pembimbing II, Gelar as Advisor II for suggestions and motivation.
3. Prof. Dr. Ir. Novirina Hendrasarie, M.T. as Dean of the Faculty of Computer Science, Universitas Pembangunan Nasional “Veteran” Jawa Timur.
4. Dr. Intan Yuniar Purbasari, S.Kom., M.Sc. as Coordinator of the Informatics Study Program, Faculty of Computer Science, Universitas Pembangunan Nasional “Veteran” Jawa Timur.

Surabaya, Month Day, Year

Nama Lengkap Mahasiswa

TABLE OF CONTENTS

LIST OF FIGURES

LIST OF TABLES

LIST OF CODE

LIST OF ABBREVIATIONS

Writing tip: Remove unused example abbreviations and sort alphabetically.

API	Application Programming Interface
CNN	Convolutional Neural Network
UI	User Interface

CHAPTER I
INTRODUCTION

1.1. Background

Complete this section: Explain the general context, ideal condition, real problem, research/practical gap, and why the study is necessary.

Write the background coherently from the broader context toward the specific undergraduate thesis problem. Use verifiable scholarly sources such as journal articles, conference papers, books, standards, or official documents [1].

Writing tip: Avoid jumping into technical details before explaining the problem. Support important claims with citations.

1.2. Problem Statement

Complete this section: Convert the research problem into specific, measurable questions that can be answered through the methodology.

1. How can an approach be designed to address the main problem in the research object?
2. How can the proposed approach be evaluated using appropriate metrics?
3. What factors influence the research results based on the analysis?

1.3. Scope and Limitations

Complete this section: Define the scope so the study remains focused. The scope may cover data, method, location, period, variables, tools, or assumptions.

1.4. Research Objectives

Complete this section: Align objectives with the problem statements. Use verifiable verbs such as analyze, design, implement, evaluate, or compare.

1.5. Research Benefits

Complete this section: Explain theoretical and practical benefits. Avoid exaggerated claims; state realistic contributions for the scale of the undergraduate thesis.

CHAPTER II
LITERATURE REVIEW

2.1. Related Work

Complete this section: Summarize the most relevant related studies. Explain their objectives, methods, data, results, and limitations. End by positioning your study against those works.

A literature review is not merely a list of article summaries. Organize it thematically so readers understand the development of knowledge, remaining gaps, and why this study is needed.

Table 2. 1. Example format for summarizing related work

Author	Focus and Method	Relevance to This Undergraduate Thesis
Author name (year)	Short summary of method, data, and main results.	Explain lessons, gaps, or distinctions for your research.

2.2. Theoretical Foundation

Complete this section: Explain concepts, theories, algorithms, standards, or frameworks that are actually used in the study.

2.2.1. First Main Concept

Write the definition, working principle, and relevance of the concept to the study. Use citations for definitions or theories that are not your own opinion.

2.2.2. Second Main Concept

Add other concepts as needed. If using formulas, explain the meaning of each symbol and why the formula is used.

2.3. Conceptual Framework

Complete this section: Explain the relationship between the problem, theory, method, data, and research outputs. A conceptual diagram can help this section.

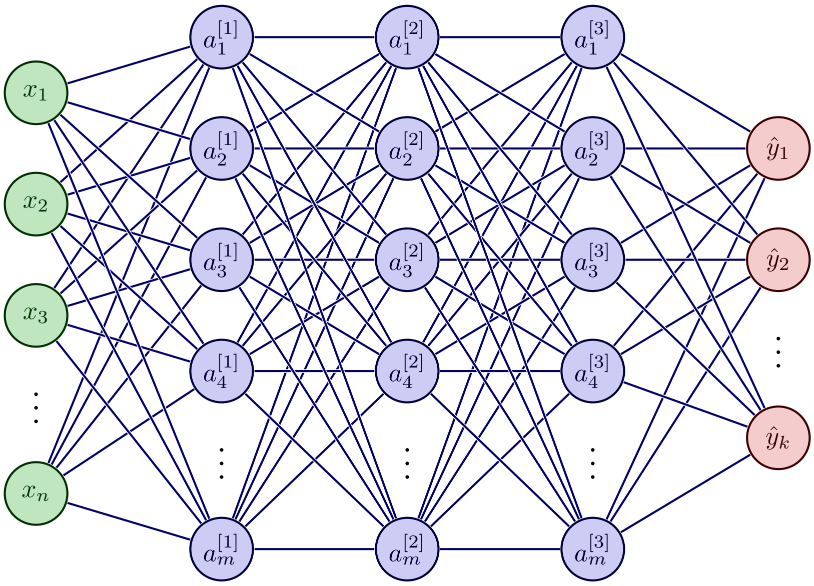

Figure 2. 1. Example figure usage in the template

An example of figure usage is shown in Figure 2. 1. Mathematical notation can be used to explain the relationship between layers in Figure 2. 1. Let the input vector be denoted as

$$\mathbf{x} = \begin{bmatrix} x_1 & x_2 & \cdots & x_n \end{bmatrix}^\mathsf{T}. \tag{2.1}$$

For the l-th hidden layer, neuron activations are computed by

$$\mathbf{a}^{[l]} = \sigma\left(\mathbf{W}^{[l]}\mathbf{a}^{[l-1]} + \mathbf{b}^{[l]}\right), \qquad l = 1, 2, \ldots, L-1, \tag{2.2}$$

where $\mathbf{a}^{[0]} = \mathbf{x}$, $\mathbf{W}^{[l]}$ is the weight matrix at layer l, $\mathbf{b}^{[l]}$ is the bias vector, and $\sigma(\cdot)$ is the activation function. The model output is denoted as

$$\hat{\mathbf{y}} = f\left(\mathbf{W}^{[L]}\mathbf{a}^{[L-1]} + \mathbf{b}^{[L]}\right), \tag{2.3}$$

with output components

$$\hat{\mathbf{y}} = \begin{bmatrix} \hat{y}_1 & \hat{y}_2 & \cdots & \hat{y}_k \end{bmatrix}^\mathsf{T}. \tag{2.4}$$

CHAPTER III
RESEARCH METHODOLOGY

3.1. Research Workflow

Complete this section: Explain the research stages from beginning to end. Ensure the workflow aligns with the problem statements and reported results.

The methodology should be detailed enough for the study to be understood and, where possible, replicated. Describe important decisions, parameters, assumptions, and reasons for method selection.

3.2. Research Object and Data

Complete this section: Describe the research object, data sources, inclusion/exclusion criteria, collection period, data quantity, and ethical or permission aspects if relevant.

3.3. Method or System Design

Complete this section: Describe the design of the method, model, system, or experimental procedure. Connect each component to the research objectives.

Code Program 3. 1: Example research algorithm format

```
Input: Research data or input
Output: Evaluated results
Perform data preparation;
Apply the proposed method;
Evaluate outputs using appropriate metrics;
Return Summary of results and analysis
```

3.4. Evaluation Scenario

Complete this section: Explain the experimental design, data split, baselines/comparators, evaluation metrics, hardware/software, and how fair comparison is ensured.

3.5. Data Analysis Technique

Complete this section: Explain how results will be analyzed: descriptive statistics, metric comparison, error analysis, user validation, or other suitable approaches.

CHAPTER IV
RESULTS AND DISCUSSION

4.1. Implementation or Research Execution Results

Complete this section: Report what was actually built, collected, or executed. Separate factual results from interpretation.

4.2. Evaluation Results

Complete this section: Present main results according to the scenarios in Chapter III. Tables should have clear captions, units, and sufficient explanation.

Table 4. 1. Example evaluation result table

Scenario	Metric 1	Metric 2
Scenario A	nilai/value	nilai/value
Scenario B	nilai/value	nilai/value

4.3. Discussion

Complete this section: Explain why the results occurred, their relationship to theory and related work, and their implications. Do not merely repeat table contents.

4.4. Research Limitations

Complete this section: State limitations honestly and specifically, such as limitations in data, time, hardware, method, validation, or user scope.

CHAPTER V
CONCLUSIONS AND RECOMMENDATIONS

5.1. Conclusions

Complete this section: Answer the problem statements directly based on Chapter IV. Do not introduce new data or claims in the conclusions.

1. The first conclusion answers the first problem statement based on the discussed results.
2. The second conclusion explains the main achievement, metric, or important finding proportionally.
3. The third conclusion may explain implications or boundaries for interpreting the results.

5.2. Recommendations

Complete this section: Provide realistic recommendations for future research or practical application. Connect recommendations to the limitations already discussed.

BIBLIOGRAPHY

[1] Nama Penulis. *Judul Buku atau Referensi Ilmiah*. Nama Penerbit, Kota, 2024.

APPENDIX

Complete this section: Use the appendix for supporting materials that are too long for the main chapters, such as research instruments, experiment configurations, interview transcripts, test evidence, or additional documentation.

Appendices should still be referenced from the main chapters. Do not include materials that are never mentioned in the manuscript.

Appendix A. Example Appendix

Write appendix content here. If the appendix contains code, data, or images, ensure sensitive information has been removed before publishing the repository.